

Supernova ZR: baseline vs. evolved

Full portfolio Pareto frontiers, seed 10009

One core; 60-minute total search budget; restarts capped at 10^6 evaluations; no max-size warmup. Baseline uses L1 loss. Evolved uses the validation-selected bundle from run 709715 and its custom affine-profile loss.

Reference family and the highlighted candidates

The task fits normalized observed flux as a function of time t (days relative to the observed peak). The 236 rows span -18.04 to 86.75 days. The accepted reference is the empirical Bazin family, not a known physical generating law:

$$F(t; A, B, C, D) = \frac{A}{B \exp(Ct) + \exp(-Dt)}.$$

The yellow boxes identify the saved review's *closest structural match*, not the lowest-loss or highest- R^2 member. B06 is the simplest exact-family match cited by the baseline review. E13 is the evolved review's near match. All 15 baseline and all 20 evolved frontier rows follow, in increasing complexity.

Baseline B06 (exact):

$$\frac{\exp(at)}{\exp(t) + b} = \frac{1/b}{(1/b) \exp((1-a)t) + \exp(-at)}, \quad a \simeq 0.955950, \quad b \simeq 2.25986 \times 10^{-5}.$$

Evolved E13 (near):

$$\frac{(0.0571477t)^3 + 0.976272}{\exp(-0.127003t) + 1.27564 \exp(0.119704t)}.$$

Its denominator has the reference structure, but its numerator contains a real nonconstant cubic term. Rounding here is for display only; the raw syntax-tree version appears in the frontier.

Metrics

Let $p_i = f(t_i)$, $S_y = \sum_i (y_i - \bar{y})^2$, and let $(\hat{a}, \hat{b})$ minimize $\sum_i (y_i - ap_i - b)^2$. Then

$$R^2 = 1 - \frac{\sum_i (y_i - p_i)^2}{S_y}, \quad R_{\text{cal}}^2 = 1 - \frac{\sum_i (y_i - \hat{a}p_i - \hat{b})^2}{S_y}.$$

The native losses are

$$L_{\text{L1}} = \frac{1}{n} \sum_i |y_i - p_i|, \quad L_{\text{custom}} = \sqrt{\text{clip}(1 - R_{\text{cal}}^2, 0, 1)} + \frac{1}{256} \frac{\sqrt{1 - R^2}}{1 + \sqrt{1 - R^2}}.$$

The custom expression matches the implementation on this dataset (its tiny target-normalization floor is inactive). Calibration fits scale and offset only, not the equation's internal constants. For constant predictions, calibration returns the target mean and $R_{\text{cal}}^2 = 0$.

Reading the tables. Loss is each method's own search loss; the two loss columns have different meanings and should not be compared numerically. Every R^2 is recomputed on the same 236 training rows, without clipping negative values. Calibrated R^2 is a diagnostic, not a held-out score or the stored raw equation's prediction quality. Constants are displayed to six significant digits; all metrics use the full-precision stored expressions. Row IDs are zero-based positions in the merged frontier, not the original per-restart PySR indices.

Reference context: [Russeil et al., Multi-View Symbolic Regression, section 5.3 and Table 4.](#)

Baseline portfolio

Native loss: mean absolute error (L1)

Seed 10009; 217 successful restarts; full merged frontier. Rows B00–B04 (1/3).

B00	Complexity 1	Loss = 0.161802 $f(t) = 0.131753$	$R^2 = -0.174873$	$R_{\text{cal}}^2 = 0$
B01	Complexity 4	Loss = 0.114846 $f(t) = \exp(t \cdot (-0.0420392))$	$R^2 = -2.39165$	$R_{\text{cal}}^2 = 0.160738$
B02	Complexity 5	Loss = 0.0874656 $f(t) = t \cdot (-0.00794238) + 0.55892$	$R^2 = 0.466576$	$R_{\text{cal}}^2 = 0.51394$
B03	Complexity 6	Loss = 0.0553266 $f(t) = \frac{325.338}{(t)^2 + 424.217}$	$R^2 = 0.790392$	$R_{\text{cal}}^2 = 0.790887$
B04	Complexity 7	Loss = 0.0455936 $f(t) = \frac{0.559926}{\exp\left((t \cdot 0.0253315)^2\right)}$	$R^2 = 0.727226$	$R_{\text{cal}}^2 = 0.727711$

Baseline portfolio

Native loss: mean absolute error (L1)

Seed 10009; 217 successful restarts; full merged frontier. Rows B05–B09 (2/3).

B05	Complexity 8	Loss = 0.045073 $f(t) = \frac{645.304}{(t)^2 + 863.234} - 0.0758309$	$R^2 = 0.762847$	$R_{\text{cal}}^2 = 0.764589$
B06	Complexity 9	Loss = 0.0255688 $f(t) = \frac{\exp(t \cdot 0.95595)}{\exp(t) + 2.25986 \times 10^{-5}}$ Closest structural match: EXACT. Algebraically in the Bazin family.	$R^2 = 0.971463$	$R_{\text{cal}}^2 = 0.972425$
B07	Complexity 10	Loss = 0.0239115 $f(t) = \frac{\sqrt{\exp(t)}}{\exp(t \cdot 0.54407) + 0.00510327}$	$R^2 = 0.975998$	$R_{\text{cal}}^2 = 0.976389$
B08	Complexity 11	Loss = 0.0229951 $f(t) = \frac{\exp(t)}{\exp(\frac{t}{0.961963}) + 1.13455 \times 10^{-5}} + (-0.0255283)$	$R^2 = 0.971604$	$R_{\text{cal}}^2 = 0.972677$
B09	Complexity 13	Loss = 0.0210628 $f(t) = \frac{\exp(t \cdot 0.328156)}{\exp(t \cdot 0.366919) + 0.0424653} + (-0.0316802)$	$R^2 = 0.976917$	$R_{\text{cal}}^2 = 0.977359$

Baseline portfolio

Native loss: mean absolute error (L1)

Seed 10009; 217 successful restarts; full merged frontier. Rows B10–B14 (3/3).

B10	Complexity 14	Loss = 0.0203803	$R^2 = 0.977902$	$R_{\text{cal}}^2 = 0.978554$
$f(t) = \frac{\exp(t \cdot (-0.0392879)) - 0.0282795}{\sqrt{\exp(-9.38494 - t) + 0.962586}}$				
B11	Complexity 15	Loss = 0.0197551	$R^2 = 0.979419$	$R_{\text{cal}}^2 = 0.979513$
$f(t) = \frac{\exp(t \cdot 0.344832)}{\exp(t \cdot 0.383571) + t \cdot (-0.00257239)} + (-0.0299226)$				
B12	Complexity 17	Loss = 0.0176212	$R^2 = 0.985718$	$R_{\text{cal}}^2 = 0.986869$
$f(t) = \frac{\exp\left((t)^2 \cdot (-0.0138108)\right) + 0.407837}{\exp\left((t \cdot (-0.0570707) + 0.972911) \cdot 0.613631\right)^2}$				
B13	Complexity 18	Loss = 0.00823029	$R^2 = 0.995682$	$R_{\text{cal}}^2 = 0.995832$
$f(t) = \frac{\exp(t \cdot (-0.0341614))}{\exp\left(-0.148379 - (t \cdot 0.06168 + (-0.47134))^3\right) - t \cdot (-0.0335344)}$				
B14	Complexity 20	Loss = 0.00805523	$R^2 = 0.995947$	$R_{\text{cal}}^2 = 0.996002$
$f(t) = \frac{\exp((0.284446 - t) \cdot 0.0350296)}{\frac{\exp((0.480282 - t \cdot 0.0617161)^3) - t \cdot (-0.0378628)}{1.14979}}$				

Source run: runs/srbench2_9-8_gt_baseline_1core_l1_portfolio_1m_no_warmup

Full-precision equations and metrics are retained in the companion JSON.

Evolved portfolio

Native loss: custom affine-profile loss

Seed 10009; 77 successful restarts; full merged frontier. Rows E00–E04 (1/4).

E00	Complexity 1	Loss = 0.701069 $f(t) = t$	$R^2 = -49147.9$	$R_{\text{cal}}^2 = 0.51394$
E01	Complexity 2	Loss = 0.700248 $f(t) = (t)^2$	$R^2 = -2.13488 \times 10^8$	$R_{\text{cal}}^2 = 0.515108$
E02	Complexity 3	Loss = 0.699418 $f(t) = t \cdot 0.00217409$	$R^2 = -0.798909$	$R_{\text{cal}}^2 = 0.51394$
E03	Complexity 4	Loss = 0.560726 $f(t) = \sqrt{t} \cdot t$	$R^2 = -49140$	$R_{\text{cal}}^2 = 0.689933$
E04	Complexity 5	Loss = 0.316638 $f(t) = \ln\left((4.23476 - t)^2\right)$	$R^2 = -976.591$	$R_{\text{cal}}^2 = 0.902123$

Evolved portfolio

Native loss: custom affine-profile loss

Seed 10009; 77 successful restarts; full merged frontier. Rows E05–E09 (2/4).

E05	Complexity 7	Loss = 0.26372 $f(t) = \ln(\exp(t \cdot (-0.281753)) + t)$	$R^2 = -256.405$	$R_{\text{cal}}^2 = 0.932377$
E06	Complexity 8	Loss = 0.178138 $f(t) = \ln(\exp(t \cdot (-0.506237)) + (t)^2)$	$R^2 = -1067.77$	$R_{\text{cal}}^2 = 0.969603$
E07	Complexity 9	Loss = 0.136175 $f(t) = \ln\left(\frac{(t)^2 + 13.7774}{t + 22.7922}\right)$	$R^2 = -202.094$	$R_{\text{cal}}^2 = 0.982437$
E08	Complexity 10	Loss = 0.132851 $f(t) = \ln\left(\left(\frac{(t)^3}{t + 16.7407}\right)^2 + 5.21675\right)$	$R^2 = -3970.93$	$R_{\text{cal}}^2 = 0.983357$
E09	Complexity 11	Loss = 0.129274 $f(t) = t \cdot (-0.0480561) + \sqrt{\sqrt{t} \cdot t + 1.97038}$	$R^2 = -360.754$	$R_{\text{cal}}^2 = 0.984234$

Evolved portfolio

Native loss: custom affine-profile loss

Seed 10009; 77 successful restarts; full merged frontier. Rows E10–E14 (3/4).

E10	Complexity 12	Loss = 0.11305	$R^2 = -1062.08$	$R_{\text{cal}}^2 = 0.988062$
$f(t) = (1.00066 - t \cdot 0.0212141)^3 + \ln\left((t)^2 - (-6.73503)\right)$				
E11	Complexity 14	Loss = 0.111361	$R^2 = -0.349908$	$R_{\text{cal}}^2 = 0.988062$
$f(t) = \left(\ln\left((t)^2 - (-6.80919)\right) + (1.00017 - t \cdot 0.0212099)^3\right) \cdot 0.0202608$				
E12	Complexity 16	Loss = 0.108949	$R^2 = -4134.8$	$R_{\text{cal}}^2 = 0.988954$
$f(t) = ((47.5941 - t) \cdot 0.026687)^3 + \ln\left((t \cdot (t)^3 + 55.6294) \cdot 0.685358\right)$				
E13	Complexity 18	Loss = 0.080043	$R^2 = 0.315679$	$R_{\text{cal}}^2 = 0.993873$
$f(t) = \frac{(t \cdot 0.0571477)^3 + 0.976272}{\exp(t \cdot 0.119704) \cdot (\exp(t \cdot (-0.246707)) + 1.27564)}$				
Closest structural match: NEAR. Extra cubic numerator; not in the Bazin family.				
E14	Complexity 20	Loss = 0.0698723	$R^2 = -9478.69$	$R_{\text{cal}}^2 = 0.995643$
$f(t) = t \cdot 0.0929013 - \left(\ln\left((-29.0947 + t)^2 + t\right) \cdot 0.599663 + \ln\left(\left((t)^3\right)^2 + 1999.67\right)\right)$				

Evolved portfolio

Native loss: custom affine-profile loss

Seed 10009; 77 successful restarts; full merged frontier. Rows E15–E19 (4/4).

E15	Complexity 22	Loss = 0.0665761	$R^2 = -9204.2$	$R_{\text{cal}}^2 = 0.996067$
$f(t) = \ln\left(\left((t)^2\right)^3 + 1999.68\right) + \ln\left(t + (t + (-28.0842))^2 + (-1.32598)\right) \cdot 0.612279 - t \cdot 0.0931809$				
E16	Complexity 23	Loss = 0.0617656	$R^2 = 0.910611$	$R_{\text{cal}}^2 = 0.996295$
$f(t) = \exp(t \cdot (-0.00890719) \cdot (t - (-3.60761))) - \frac{-0.920096 - t \cdot 0.0892947}{\exp(1.31416 - t \cdot t \cdot (-0.00104235))}$				
E17	Complexity 25	Loss = 0.0501894	$R^2 = -11076.5$	$R_{\text{cal}}^2 = 0.997854$
$f(t) = \frac{\ln\left(t \cdot 4.54376 + (t + (-26.074))^2\right)}{t \cdot 0.00108345 - (-0.193845) - \frac{-27.5708}{\frac{0.831265 - t}{-0.19307 - t}}}$				
E18	Complexity 27	Loss = 0.0495817	$R^2 = -0.895537$	$R_{\text{cal}}^2 = 0.997761$
$f(t) = \frac{\ln\left(\left((25.3662 - t)^2 + 49.077\right) \cdot 63.3788\right)}{\left((t)^2 + 120.801\right) \cdot \left(\frac{63.3803}{t + t \cdot 2.41807 + (-2.27579)}\right)^2 + t}$				
E19	Complexity 29	Loss = 0.0492654	$R^2 = -0.0573048$	$R_{\text{cal}}^2 = 0.997764$
$f(t) = \frac{\ln\left(\left((25.3699 - t)^2 + 49.0231\right) \cdot 63.3676\right)}{\left((t)^2 + 120.822\right) \cdot \left(\frac{63.3802}{t + t \cdot 2.41698 + (-2.28243)}\right)^2 + t} - (-0.2059)$				

Source run: runs/709715/srbench2_9-8_ground_truth_1core_portfolio_1m_no_warmup

Full-precision equations and metrics are retained in the companion JSON.